

Assessment of trace metals and porphyrins in excreta of Humboldt penguins (*Spheniscus humboldti*) in different locations of the northern coast of Chile

José E. Celis · Winfred Espejo · Daniel González-Acuña · Solange Jara · Ricardo Barra

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Abstract To add data on trace metal contamination of Humboldt penguins in the South Pacific, levels of trace metals (As, Hg, Pb, Cu, Zn, and Cd) and porphyrins (copro-, uro-, and proto-) in excreta of Humboldt penguins that inhabit some important nesting sites on the northern coast of Chile were determined. Fresh excreta were collected on Pan de Azúcar Island, Chañaral Island, and Cachagua Island, from December 2011 to January 2012. Concentration of metals was determined by flame atomic absorption spectrophotometry, whereas porphyrins levels were measured by fluorimetric analysis. Concentrations (dry weight) of Cu ($199.67 \mu\text{g g}^{-1}$), As ($7.85 \mu\text{g g}^{-1}$), and Pb ($12.78 \mu\text{g g}^{-1}$) were higher ($p \leq 0.05$) in Cachagua Island. Colonies from Pan de Azúcar Island showed the highest levels of Hg ($0.76 \mu\text{g g}^{-1}$), Cd ($47.70 \mu\text{g g}^{-1}$), and Zn ($487.10 \mu\text{g g}^{-1}$). Samples from Cachagua Island showed the highest ($p \leq 0.05$) levels of copro- (2.16 nmol g^{-1}), uro- (2.20 nmol g^{-1}), and protoporphyrins (2.23 nmol g^{-1}). There was a positive correlation between the metals As, Pb, and Cu with uro-, copro-, and protoporphyrins. The results indicated that penguin colonies from Cachagua Island are more exposed to metal contamination than penguin colonies from Pan

de Azúcar and Chañaral Islands, thus being more likely to develop certain diseases caused by contamination with metals. Considering biomagnification, the metals detected in the excreta of Humboldt penguins can be a source of contamination from marine environments to terrestrial ecosystems, which could also affect other living organisms.

Keywords Heavy metals · Seabirds · Penguins · Excrement · Biomonitoring · Sea pollution

Introduction

The Humboldt penguin is a species of seabird that naturally inhabits the coast between Peru (5° S) and Chile (43° S), closely linked to the Humboldt Current, from which its name has been taken (Araya and Millie 2000). This bird is vulnerable and although it is legally protected, it is still hunted both for food and bait (Ellenberg et al. 2006). The Humboldt penguin is a bird at the top of the food chain of marine ecosystems, thus playing an important role in the ecology of the coastal zones of northern Chile (Araya and Millie 2000). There is some evidence indicating that Humboldt penguins are particularly vulnerable to human activities, which become a source of behavioral and physiological stress that alters the breeding success of these birds (Ellenberg et al. 2006). Anthropogenic activities constitute an important source of trace metals (Boersma 2008). The northern Chilean coast is continuously affected by many human actions such as mining, urban centres, local

J. E. Celis (✉) · W. Espejo · D. González-Acuña
Facultad de Ciencias Veterinarias,
Universidad de Concepción,
Casilla 537, Chillán, Chile
e-mail: jcelis@udec.cl

S. Jara · R. Barra
Unidad de Sistemas Acuáticos, Centro de Ciencias
Ambientales EULA-Chile, Universidad de Concepción,
Casilla 160-C, Concepción, Chile

fishing industries, seaports, and roads for the transit of heavy vehicles and airports (Ellenberg et al. 2006). Birds, like mammals, are useful indicators of the degree of contamination by trace metals in the environment because of their wide distribution and for being top predators (De Moreno et al. 1997).

Heavy metal contamination is widespread globally as a result of anthropogenic activities (Nriagu and Pacyna 1988). Accidental oil spills, sewage, hazardous wastes, pesticides, mining, smelting, or forest fires can contribute to increased levels of Hg, As, and Cd levels, whereas electrical devices, mining, or explosives can increase Pb levels in marine ecosystems (Boersma 2008). Some metals, such as Cu and Zn at low levels are essential for the physiological and biochemical functions of living organisms (Tahri et al. 2005). However, all metals can be potentially toxic and even carcinogenic, depending on the levels in blood or tissues and the chemical nature of the metal (Bengtsson et al. 2006; Miller et al. 2004). Some evidence indicates that trace metals are toxic for birds and can cause severe damage to kidneys, liver, and the central nervous system (Kim et al. 1998; Horai et al. 2007). Considering the cumulative nature and persistence of metals, these are found not only in water, air, or soil, but they are also found in both flora and fauna (Zhao et al. 2006). The effect of chemical pollutants, such as trace metals, is more severe in species that dominate the higher levels of the food chain because these pollutants tend to bioaccumulate (Burger et al. 2000).

The risk of chemical exposure in both wildlife and humans can be achieved by using porphyrins as a marker of biochemical effects (De Matteis and Lim 1994). Porphyrins are produced and accumulated in trace amounts in erythropoietic tissues such as the liver and kidneys and excreted via urine or feces (Lim 1991). Porphyrins are capable of binding metals, and they can be detected in different biological materials, even at very low concentrations (De Matteis and Lim 1994). This characteristic makes it possible for porphyrins to be used as biomarkers for measuring the level of exposure to trace metals in seabirds (Casini et al. 2003). In addition, some trace metals can interfere with heme biosynthesis causing alterations in the level of porphyrins that are accumulated or excreted (Casini et al. 2001). The resulting oxide by-products as copro-, uro-, and protoporphyrins are not toxic at normal levels, but when accumulated in excess can affect liver and bone marrow (Lim 1991). There is evidence indicating a direct correlation between metal pollution

and accumulation of porphyrins in the liver and excreta (Leonzio et al. 1996).

There are many ways of measuring the degree of contamination of seabirds, such as blood, egg shells, feathers, bones, and internal organs (Borg et al. 1966; Ninomiya et al. 2004). However, these techniques of biomonitoring are usually very invasive for birds, and most of them are restricted by the International and National Regulations for Wildlife Protection. Moreover, seabirds are significant biovectors of contaminants from the ocean to the land (Michelutti et al. 2009). For that reason, excreta have been used as an appropriate bioindicator for the assessment of trace metals in penguins to study pollution in marine ecosystems with a minimum of human intervention (Sun and Xie 2001; Ancora et al. 2002; Yin et al. 2008; Celis et al. 2012).

The huge development of mining in northern Chile has adversely affected coastal ecosystems by high concentrations of metals (Salamanca et al. 2004; Ramirez et al. 2005; Stauber et al. 2005). Pollution by metals can also occur in the form of antifouling paintings, sewage, waste incineration, coal combustion, oil spills, pipe corrosion, and solid wastes (Salamanca et al. 2004; Santos et al. 2005; Bargagli 2008). In northern Chile, many industrial activities discharge their wastes into the sea, thus some chemicals have increased their levels to those considered as dangerous to ecosystem health (Vermeer and Castilla 1991; Zúñiga 1998; Cortés and Luna-Jorquera 2011). Despite the potential impacts posed to seabirds by metal contamination, there is a lack of research on possible contamination in Humboldt penguins. Consequently, the objectives of the study were: (1) to determine the levels of Hg, Cd, Pb, As, Cu, and Zn in excreta of Humboldt penguins that inhabit some important nesting sites on the northern coast of Chile and (2) to assess the use of porphyrins as a biomarker of metal contamination in Humboldt penguins.

Materials and methods

Nesting sites and collection of excreta samples

In Chile, the population of Humboldt penguins is between 40,000 and 47,000 individuals, which mostly nest on 35 sites along the coast (Schlosser et al. 2009). Between December 2011 and January 2012, fresh droppings of Humboldt penguin colonies were collected from three of the most important nesting sites in the northern of Chile,

as shown in Fig. 1: (1) Pan de Azúcar Island (26°09' S, 70°40' W), (2) Chañaral Island (29°01' S, 71°34' W), and (3) Cachagua Island (32°35' S, 71°27' W).

Pan de Azúcar Island is an oval island (1.5×1 km), which is located 1 km from the coastline and 18 km distant from Chañaral Bay, a highly impacted area as a consequence of mining activities that occurred between 1950 and 1960. Pan de Azúcar Island has one of the largest Humboldt penguin colonies in northern Chile, supporting approximately 5,000 individuals. Chañaral Island is a circular island (2 km radius) located 7 km from the coastline and 100 km north of Coquimbo Bay. It is an isolated area which supports the largest

Humboldt penguin colonies in northern Chile, with approximately 22,000 individuals and is visited by tourists. Conversely, Cachagua Island is a small island (250×200 m) located 100 m off the coastline next to a local town and 20 km from Quintero Bay, a very industrialized area. Even though Cachagua Island is a small island, it constitutes an important nesting site for the Humboldt penguins, and supports a population of approximately 2,000 individuals (Meza et al. 1998).

Samples were carefully taken from the ground to avoid contamination. Excreta samples were always handled with disposable plastic gloves. Penguin droppings were taken with plastic spatulas from the top of the mass of excrement

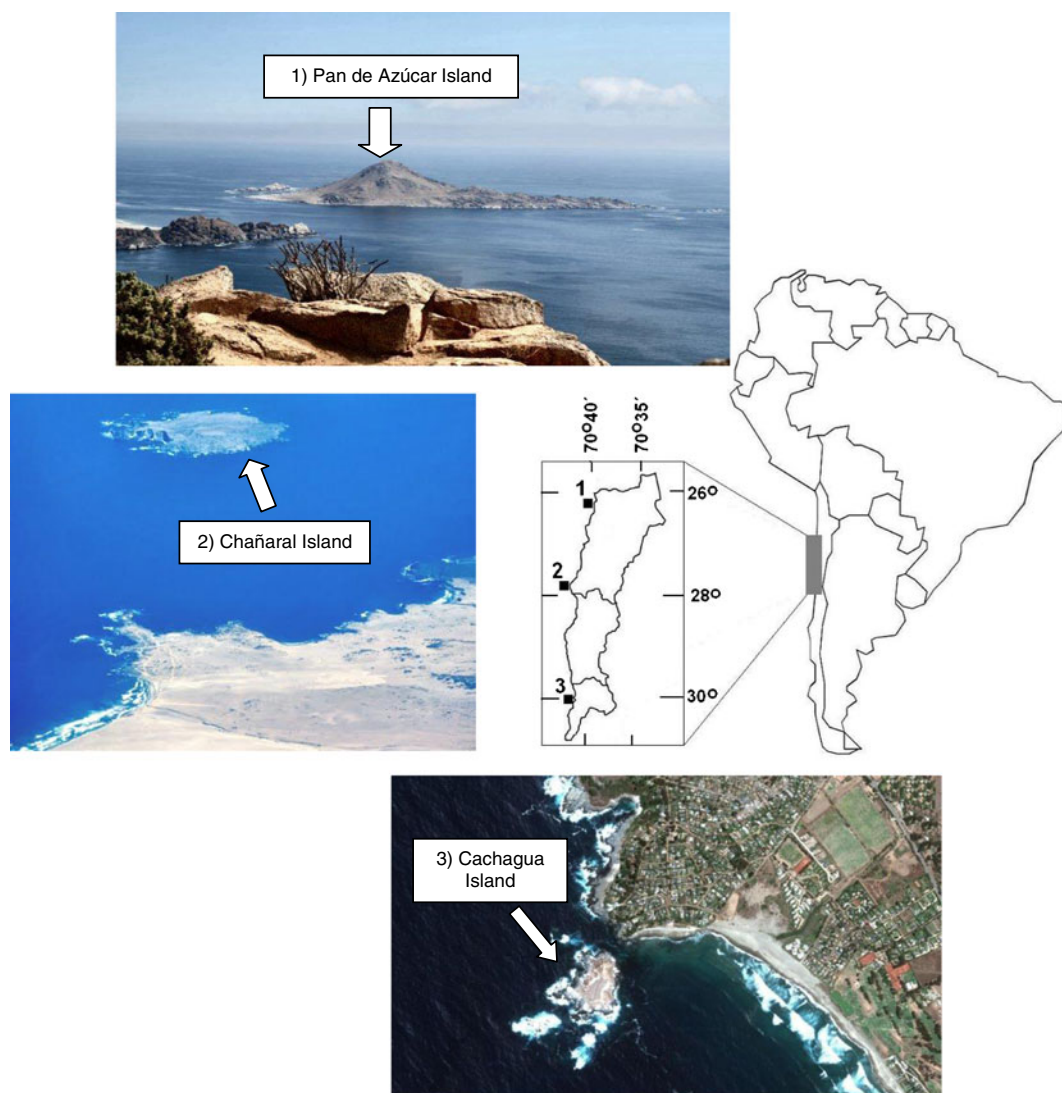


Fig. 1 Location of sampling sites on the northern coast of Chile

deposited on the ground to avoid contact with any contaminant deposits in the soil. Each excrement sample was collected from many individuals, spanning the length of the penguin colony as it was not possible to evaluate metal contamination in each individual, but rather only in a group of that species. Very clean steel containers and sealed plastic bags were also used. Once brought to the laboratory, all samples were kept in sealed plastic bags and stored in pre-cleaned aluminum foil at -4°C for transport until their analyses.

Preparation of samples and chemical analysis

In the laboratory (accredited ISO 17025), samples were washed with distilled water (Milli-Q system, Millipore, USA), dried at room temperature and then ground and screened (24 mesh dm^{-2}). For Pb, Cu, As, Cd, and Zn, samples were digested with nitric acid, hydrochloric acid, and perchloric acid. For Hg, samples were digested with nitric acid and 30 % H_2O_2 (6:5, v/v). The levels of elements in the samples were determined by atomic fluorescence for Hg, atomic absorption spectrometry for As, and atomic absorption spectrometry with graphite furnace for Cd and Pb. Levels of Cu and Zn were determined using flame atomic absorption spectrophotometry. To ensure quality control, a certified reference material, human hair, was used as an internal standard in a proportion of 10 % for each batch of samples.

Porphyrins were determined as described by Lockwood et al. (1985). Briefly, 1 ml hydrochloric 5 N solution was placed in a graduated centrifuge tube containing 100 mg excreta and vortex-mixed. Diethyl ether (3 ml) was added and mixed to obtain an emulsion; then, 3 ml water were added to the solution and mixed again, after which it was centrifuged at $700\times g$ for 10 min. The supernatant was

decanted, and the precipitant containing the porphyrins was exposed to fluorimetric analysis (Casini et al. 2001).

Statistical analysis

The detected levels are presented as mean $\pm$ standard deviation in dry weight. Nonparametric statistical methodologies were used because of the assumptions of normality and homoscedasticity were not met even after the log transformation of the data. Differences among metal and trace concentrations were assessed by means of the Kruskal–Wallis nonparametric analysis of variance and Mann–Whitney U tests. Post hoc tests were carried out by means of Kruskal–Wallis analyses, using the critical differences of mean rank. Spearman rank correlation coefficients were calculated between metal and trace levels. A significance level of $p < 0.05$ was considered to indicate statistical significance. Data were analyzed by through the SPSS 15.0 statistical software

Results

Geographical differences

Concentrations of As, Cd, Cu, Hg, Pb, and Zn in excreta of *Spheniscus humboldti* are shown in Table 1. The excreta of Humboldt penguins showed significant differences among the different locations for several trace metals (Table 2). Levels of As, Pb, and Cu in excreta of this species are significantly higher in Cachagua Island in comparison to Pan de Azúcar Island and Chañaral Island. With respect to Cd and Cu, *S. humboldti* droppings showed that the levels in Pan de Azúcar Island are

Table 1 Trace metal concentrations (micrograms per gram dry weight) in the excreta of Humboldt penguin colonies that inhabit three important nesting sites on the northern coast of Chile

Heavy metals	Sampling locations		
	Pan de Azúcar Island ($n=20$)	Chañaral Island ($n=19$)	Cachagua Island ($n=24$)
Cu	147.79 $\pm$ 146.42	69.62 $\pm$ 24.98	199.67 $\pm$ 81.78
As	1.84 $\pm$ 2.65	0.36 $\pm$ 0.40	7.86 $\pm$ 4.88
Pb	1.80 $\pm$ 3.00	1.59 $\pm$ 2.12	12.79 $\pm$ 9.97
Cd	47.70 $\pm$ 38.71	21.24 $\pm$ 18.35	42.47 $\pm$ 45.55
Hg	0.77 $\pm$ 0.83	0.46 $\pm$ 0.19	0.61 $\pm$ 0.40
Zn	487.11 $\pm$ 395.15	222.55 $\pm$ 59.20	0.83 $\pm$ 0.33
Mean values $\pm$ standard deviation			

significantly higher than those in Chañaral Island. There was more Zn in Pan de Azúcar Island and Chañaral Island than in Cachagua Island, whereas the levels of Zn did not show significant differences between Pan de Azúcar Island and Cachagua Island. Levels of Hg did not show significant differences among the studied locations, whereas levels of Cd did not show significant differences between Pan de Azúcar Island and Cachagua Island.

The levels of copro-, uro-, and protoporphyrins are shown in Table 3. The highest significant concentrations were detected in excreta of *S. humboldti* at Cachagua Island (Table 4). Levels of porphyrins did not show significant differences between Pan de Azúcar Island and Chañaral Island.

Correlation between metals and porphyrins

Several significant positive correlations were observed between trace metals and porphyrins (Table 5). A positive correlation ($p \leq 0.05$) between As, Pb, and Cu with copro-, uro-, and protoporphyrins was noted. With regard to Zn, there was a negative correlation ($p \leq 0.05$) with the porphyrins measured. The metals Hg and Cd did not show significant correlation with any of the porphyrins studied.

Discussion

The lack of published data on trace metal and porphyrin levels in excreta of Humboldt penguins makes it difficult to discuss these results. For that reason, it was necessary to compare the results of the current study with those data obtained from studies performed on penguins at other locations and birds phylogenetically similar to penguins.

The levels of metals and trace elements found in excreta of Humboldt penguins, considering all the sampling sites together, indicated the following

relation: $Zn > Cu > Cd > Pb > As > Hg$. This relation is similar to that reported by Metcheva et al. (2011) in feces of Gentoo penguins (*Pygoscelis papua*) from Antarctica: $Zn > Cu > As > Cd > Pb$. The same descendent order among Cu, Zn, and Pb was observed by Yin et al. (2008) in Chinstrap penguin (*Pygoscelis antarctica*) droppings from King George Island, Antarctica. The lower Pb levels found in penguin excreta can be explained because this metal tends to bioaccumulate mainly in bones (Teodorova et al. 2003).

The highest mean Cd levels measured in Humboldt penguin excreta ($47.70 \mu\text{g g}^{-1}$, dry weight) were found at Pan de Azúcar Island. These values were about nine times higher than those levels observed by Ancora et al. (2002) in feces of Adélie penguins (*Pygoscelis adeliae*) from Antarctica. In our study, the Cd levels were 28 and 46 times higher than those found by Celis et al. (2012) and Metcheva et al. (2011) in excreta of Gentoo penguins from Antarctica Peninsula, respectively. Cadmium is a metal that has acquired a great ecotoxicological importance as it is directly associated with human activities (Boersma 2008). When Cd is taken from its insoluble natural deposits then this chemical is widely distributed into air, water, and soil, with water being the most important site for its final deposition as a soluble form (Nriagu and Pacyna 1988). This is crucial because once Cd is deposited in waters it passes to the food chain (Bargagli et al. 1996). The main sources of Cd are mining, fossil fuels combustion and P fertilizers (Duruibe et al. 2007).

With regard to Pb, the highest mean levels of this element were found in excreta of Humboldt penguins ($12.79 \mu\text{g g}^{-1}$, dry weight, at Cachagua Island) were 17 times higher than those reported by Celis et al. (2012), 32 times the levels measured by Metcheva et al. (2011), and 13 times those reported by Yin et al. (2008), all data collected from feces of Gentoo penguins at Antarctica. In addition, our Pb levels overcame by about 32 times the

Table 2 Significant differences in trace metal concentrations among the studied locations

Location	Pan de Azúcar Island	Chañaral Island	Cachagua Island
Pan de Azúcar Island		Cd ($H_{2.63}=9.04$)** Cu ($H_{2.63}=22.38$)*	Zn ($H_{2.63}=44.64$)*
Chañaral Island			Zn ($H_{2.63}=44.64$)*
Cachagua Island	As ($H_{2.54}=31.30$)* Pb ($H_{2.62}=36.81$)* Cu ($H_{2.63}=22.38$)*	As ($H_{2.54}=31.30$)* Pb ($H_{2.62}=36.81$)* Cu ($H_{2.63}=22.38$)*	

* $p \leq 0.01$, level of significance;

** $p \leq 0.0001$, level of significance

Table 3 Porphyrin concentrations (nanomoles per gram dry weight) in excreta of Humboldt penguin colonies that inhabit three important nesting sites on the northern coast of Chile

Porphyrins	Sampling locations		
	Chañaral Island (n=20)	Pan de Azúcar Island (n=19)	Cachagua Island (n=24)
Copro-	0.32±0.48	0.61±0.76	2.16±1.57
Uro-	0.37±0.48	0.58±0.70	2.20±1.65
Proto-	0.25±0.35	0.39±0.45	2.23±1.94

Mean values±standard deviation

values reported by Ancora et al. (2002) in excreta of Adélie penguins. Pb is released by human activities such as burning fossil fuels, solid waste incineration, paints, and accidental oil spills (Bargagli 2008). Our Pb levels appear to be higher than $10 \mu\text{g g}^{-1}$ (dry weight) as described in feces from other species of birds from non-contaminated areas (Beyer et al. 1997; Mateo et al. 2006; Martinez-Haro et al. 2010), thus suggesting that some adverse physiological effects could be triggered in Humboldt penguins monitored in northern Chile.

The highest mean levels of Hg found ($0.77 \mu\text{g g}^{-1}$, dry weight, measured at Pan de Azúcar Island) were about eight times higher than those levels reported by Yin et al. (2008) and ten times lower than those measured by Celis et al. (2012) in feces of Gentoo penguins. However, Hg levels found in this study were three times higher than those found by Ancora et al. (2002) in the excreta of Adélie penguins. Several human activities are known to be Hg sources, such as coal combustion, waste incineration, paints and oil spills (Bargagli 2008). In addition, volcanic emissions release Hg into the atmosphere, resulting in the deposition of airborne Hg to the marine ecosystem as a natural event (Bargagli et al. 1998).

In this study, the highest mean As levels ($7.86 \mu\text{g g}^{-1}$, dry weight) were found at Cachagua Island. These values were around 16 times higher than those reported by Celis et al. (2012) and 1.5 times higher than those levels measured by Metcheva et al. (2011), in excreta of Gentoo penguins. In general, little

information has been reported about As levels in excreta of penguins. Arsenic is in the water, which is absorbed by krill and hence accumulates in the food chain, passing to top predators such as penguins (Metcheva et al. 2011). Comparing our As levels with those from other species, the feces of Humboldt penguins contain 3.5 times higher As levels than in found in the excreta of southern giant petrels, and 5 times more than in feces of the Antarctic seals (Xie and Sun 2008). Arsenic is released into the atmosphere as dust from mining processes, and particularly from copper, gold, and lead smelters and refineries (Durube et al. 2007).

Regarding to Cu, the maximum mean levels found in our study ($199.67 \mu\text{g g}^{-1}$, dry weight, at Cachagua Island) were twice as higher as those levels measured by Metcheva et al. (2011) in Gentoo penguin excreta, and two times lower than those reported by Yin et al. (2008) in feces of Adélie penguins. Copper is released into the environment mainly as a consequence of mining and smelter processes (Durube et al. 2007).

With regard to Zn, the highest mean levels found ($487.11 \mu\text{g g}^{-1}$, dry weight, at Pan de Azúcar Island) were 3.4 times higher than levels measured by Metcheva et al. (2011) in excreta of Gentoo penguins and 1.3 times higher than those reported by Yin et al. (2008) in Adélie penguin droppings. In the current study, the maximum Zn concentrations are directly related to the highest Cd and Hg levels in droppings of Humboldt penguins at Pan de Azúcar Island. The Zn

Table 4 Significant differences in porphyrin concentrations between the studied locations

Location	Pan de Azúcar Island	Chañaral Island	Cachagua Island
Pan de Azúcar Island			
Chañaral Island			
Cachagua Island	Copro- ($H_{2,62}=27.03$)*	Copro- ($H_{2,62}=27.03$)*	
	Uro- ($H_{2,62}=24.49$)*	Uro- ($H_{2,62}=24.49$)*	
	Proto- ($H_{2,62}=25.65$)*	Proto- ($H_{2,62}=25.65$)*	

* $p \leq 0.0001$, level of significance

Table 5 Spearman correlation between trace metals and porphyrins found in *Spheniscus humboldti* droppings

	As	Hg	Pb	Cd	Cu	Zn	Copro-	Uro-	Proto-
As	1								
Hg	0.162	1							
Pb	0.713*	0.171	1						
Cd	0.081	0.537*	−0.060	1					
Cu	0.579*	0.476*	0.502*	0.438*	1				
Zn	−0.522*	0.317*	−0.572*	0.358*	−0.183	1			
Copro-	0.594*	−0.175	0.421*	−0.017	0.374*	−0.529*	1		
Uro-	0.577*	−0.191	0.414*	−0.079	0.320*	−0.546*	0.973*	1	
Proto-	0.594*	−0.157	0.427*	−0.061	0.338*	−0.546*	0.969*	0.994*	1

* $p \leq 0.05$, level of significance; $n = 63$

is a trace metal essential for biota, and it is directly linked to Hg and Cd, in such a way that high levels of Zn may be due to reactive processes of physiological adaptation of penguins to high concentrations of Cd and Hg (Metcheva et al. 2006).

There was a positive correlation between porphyrins (uro-, copro-, and proto-) and some metals (As, Pb, and Cu). There is some evidence indicating a high affinity between Pb and Hg levels with porphyrins in excreta of Gentoo penguins (Celis et al. 2012), which could be linked to physiological problems in the pancreas and kidneys of birds (Casini et al. 2003). This is to say that colonies of Humboldt penguins that inhabit in Cachagua Island, which showed the maximum porphyrin and Pb levels in this study, could possibly receive negative effects from heavy metal contamination. In the case of Zn, there was a negative correlation with porphyrins, while Cd and Hg showed no correlation with any of the porphyrins studied. In other words, unlike Pb, porphyrins cannot be used as reliable biomarkers for the detection of Zn levels in excreta of Humboldt penguins. The reason for this is that Zn is linked with protoporphyrin in the presence of high concentrations of Pb, which is why Zn is more detected in blood than in droppings (Cárdenas-Bustamante et al. 2001).

The highest levels of nonessential metals as Pb, As and Cu in excreta of Humboldt penguins at Cachagua Island, could be related to anthropogenic activities in the surrounding area. This small island is located next to a local town. In fact, there is a heavy concentration of industries at Quintero Bay, distant 20 km south of Cachagua Island, which may contribute to the pollution of coastal environments. In this area, there are more than

ten different types of industries such as coal power plants, lubricants, cement, oil/gas refineries, chemicals, seaports, and heavy vehicle traffic (CPL 2011). In this area, one of the most important copper refineries of Chile is located, which produces 450,000 t of copper cathodes, bars of gold and silver shotblasts yearly, with its emissions being discharged into the sea and atmosphere (González et al. 2008). Additionally, Quintero Bay has an operational power station which consumes annually 700,000 t of coal (Sánchez et al. 1999).

Conversely, the highest Cd and Hg levels detected in excreta of Humboldt penguins at Pan de Azúcar Island could be linked to mining wastes which were deposited in Chañaral Bay between 1959 and 1989. In addition, Chañaral Bay, 18 km south from Pan de Azúcar Island, has human activities such as a town with a population of about 30,000 people, as well as seaports, a sulfuric acid plant, and heavy traffic related to mining activities, which may explain these higher Cd and Hg levels in excreta of Humboldt penguins living at Pan de Azúcar Island.

Cachagua Island and Pan de Azúcar Island are both located very close to the coastline and at a very short distance north from sites with much anthropogenic activity. The sea currents of northern Chile typically flow north along the coast from the south, following the same pattern of the local winds (Schneider et al. 2003). Metals can be transported as particles in the air (Berg et al. 2004). Thus, some pollutants possibly are arriving at these islands by water and air. By contrast, lower levels of metals found in Chañaral Island are due to the total absence of any industries, with only some small human settlements in the surrounding area. The

Chañaral Island is located far from the coastline and more than 100 km north from Coquimbo Bay, a site with human activities, which might explain why penguins here are less exposed to metals.

There is evidence indicating high concentrations of Cu, Pb, and Zn in the water column as well as in the tissues of bivalves on the coast of the North of Chile (Salamanca et al. 2004). Another study indicated that As ($4.11 \mu\text{g g}^{-1}$) and Cu ($29.60 \mu\text{g g}^{-1}$) levels found in the liver of fish living at same latitudes exceeded the maximum limits ($1 \mu\text{g g}^{-1}$ for As and $10 \mu\text{g g}^{-1}$ for Cu) acceptable for human consumption (Montenegro and González 2012).

Our mean levels of trace metals detected in excreta of Humboldt penguins are explained by the potential of bioaccumulation and biomagnification of these chemical elements. These metals can be transported via excreta from marine environments to terrestrial ecosystems. This assumption is based on some studies showing that excreta of penguins increased the concentration of As (Xie and Sun 2008) and Hg (Nie et al. 2012) in sediments collected at Antarctic marine ecosystems and terrestrial environments, respectively. Sediments are extremely important to the food web and serve as a reservoir of metals for bioaccumulation and trophic transfer (Burton 2002). Other studies performed in gull faeces from the northern hemisphere have shown a strong soil metals enrichment with Cd, Cr, and Zn (Otero-Perez 1998) and Cu, Pb, and Zn (Headley 1996) at the most superficial depths and close to the nests of seabirds. A recent study showed that even on remote areas penguin excrement can increase Pb in Antarctic soils up to 5.19 mg kg^{-1} (Perfetti-Bolaño and Moreno 2013). As penguins feed high on the marine food web, they can biomagnify contaminants, thus transferring bioaccumulated metals into specific receptor sites at concentrations far beyond those transported by abiotic processes (Blais et al. 2007). This may have serious implications for any biota and possible incorporation into the terrestrial food web. Evidence on sediment quality shows that the incidence of benthic macroinvertebrate effects increase when concentrations of $\text{As} > 8.2$, $\text{Cd} > 1.2$, $\text{Cu} > 34$, $\text{Pb} > 46.7$, $\text{Hg} > 0.15$, and $\text{Zn} > 150 \mu\text{g g}^{-1}$ (Burton 2002). Our As, Cd, Pb, Hg, Cu, and Zn levels detected are above threshold values for these metals, indicating that Humboldt penguins are biotransporting metals from the sea to the islands to toxic levels for the surrounding biota.

As trace metals are highly toxic to animals, and because As and Cd both can produce cancer, neurological

disorders by Hg, or damage to the brain and kidneys as a consequence of Pb contamination (Duruibe et al. 2007), our findings suggest the need for tracking of non essential metals, particularly in those species used for human consumption, such as fish and squid, both species being important in the diet of Humboldt penguins (Herling et al. 2005).

Conversely In conclusion, the present work represents the data on the trace metals and porphyrins in excreta samples of Humboldt penguin colonies collected from three nesting sites at northern Chile. The levels of As, Cd, Cu, Hg, Pb, and Zn showed great variations among the studied sites, as well as copro-, uro-, and protoporphyrins. Humboldt penguin colonies nesting in areas close to anthropogenic sources, Cachagua and Pan de Azúcar Islands, showed higher levels of chemicals than those of Chañaral Island. Levels of metals in excreta of Humboldt penguins are higher than levels of these metals detected in droppings of other penguins from Antarctica. There was a positive correlation among As, Pb, and Cu with porphyrins, which implies that penguin colonies at Cachagua Island are more able to develop diseases as consequence of these metals. The penguin colonies studied showed a clear capacity to accumulate and eliminate metals and porphyrins in the excreta. Given the bioaccumulation of metals in seabirds occupying highest position in the food chain, Humboldt penguins, like most biovector organisms can transfer bioaccumulated metals into terrestrial ecosystems via excreta that could be affecting some important living organisms of northern Chile. The study of penguin feces is interesting for monitoring trace metals and porphyrins as they are directly associated with human activities typically present in northern of Chile.

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